llm-guard-kit

v0.17.0 Architecture, Usage Guide & API Reference

Real-time confidence scoring for LLM agents — AUROC 0.817 at $0 inference cost

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llm-guard-kit is an open-source library for real-time reliability monitoring of multi-step LLM agents. It detects failures, assigns confidence tiers, and routes to recovery adapters — all with validated AUROC scores reported on held-out evaluation sets.

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1 Executive Summary

Problem

Multi-step LLM agents using ReAct-style reasoning (think → search → observe → repeat) fail silently. When a search returns empty results, the agent loops, guesses, or confidently produces a wrong answer. There is no built-in mechanism to detect these failures before they propagate to users or downstream agents.

Solution

llm-guard-kit monitors every reasoning chain in real time. It computes a `risk_score` $\in [0, 1]$ by combining:

- **SC_OLD behavioral signals** ($0, 19 ms): 12 structural features extracted from the chain (loop rate, empty-observation rate, repeated-query rate, step count, etc.) fed into a fitted logistic-regression ensemble.
- **MiniJudge** ($0): a LogReg distilled from Sonnet judge decisions, shipped as a pre-trained `.pkl` inside the package (AUROC 0.747).
- **P(True) via Haiku** (∼$0.0003): prompts Haiku to rate answer correctness on a 1-5 scale, domain-agnostic.
- **Sonnet judge** (∼$0.007): full-chain GOOD/BORDERLINE/POOR verdict for highest cross-domain precision.

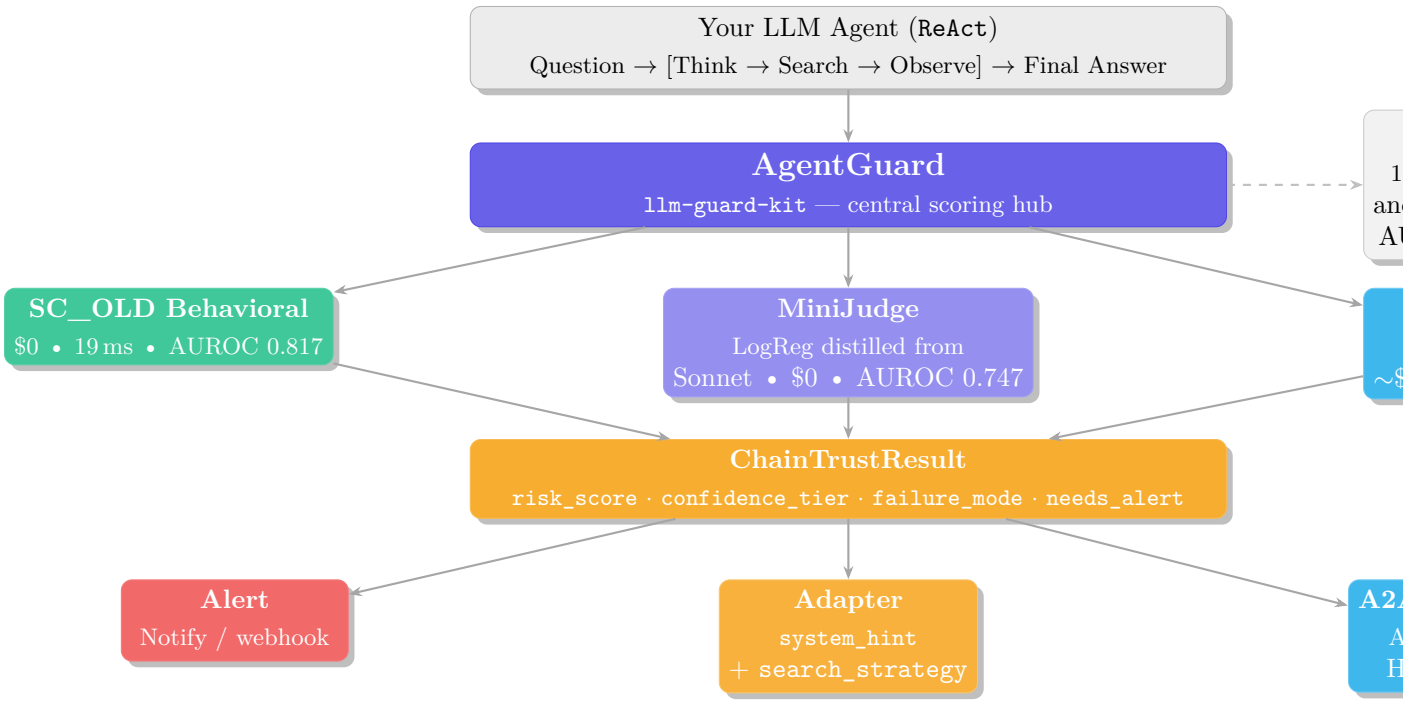
Key Results

Signal	Within-Domain AUROC	Cross-Domain AUROC	Cost/Chain
MiniJudge (behavioral LogReg)	0.747 ± 0.10	—	$0
SC_OLD behavioral ensemble	0.817	0.703 (2Wiki) / 0.659 (TV)	$0
SC_OLD + P(True) (Haiku)	—	0.74	∼$0.0003
SC_OLD + Sonnet judge	0.777	0.741	∼$0.007
Alert precision @ FPR $\leq$ 10%	0.908	—	—

Table 1: Validated performance on held-out evaluation sets. Within-domain = HotpotQA (200 chains). Cross-domain = TriviaQA n=1000, 2WikiMultiHop n=200.

2 System Architecture

The figure below shows the layered architecture of `llm-guard-kit`.



3 Component Reference

3.1 MiniJudge — $0 Local Judge

What it is: A pre-trained `sklearn` pipeline (StandardScaler + LogisticRegression on 11 SC_OLD features) distilled from Sonnet judge verdicts. Shipped as a `.pkl` inside the package.

When to use: Drop-in replacement for Haiku/Sonnet when API access is unavailable or cost is constrained. Ideal for CI/CD pipelines and batch evaluation.

Performance: AUROC 0.747 ± 0.10 (5-fold CV, HotpotQA 200 chains, exp159). Cost: $0.000/chain.

```
from llm_guard import MiniJudge

judge = MiniJudge()            # auto-loads bundled pkl
risk  = judge.score(question, steps, final_answer)  # -> float [0,1]

# Score using alias (drop-in with AgentGuard)
risk  = judge.score_chain(question, steps, final_answer)

# Retrain on your own labeled data
judge.fit(labeled_chains)      # each chain needs "correct" bool field
```

3.2 AgentGuard (Behavioral) — Production Baseline

What it is: Behavioral SC_OLD scoring using 12 structural features extracted from the ReAct chain. Fitted logistic-regression ensemble with no API dependency.

When to use: Default choice for production. $0 cost, 19 ms p50 latency. AUROC 0.817 within HotpotQA, 0.659–0.703 cross-domain.

Performance: AUROC 0.817 within-domain; alert precision 0.908 at FPR $\leq 10\%$ (exp92 conformal calibration).

```
from llm_guard import AgentGuard
```

```

guard  = AgentGuard()   # no API key needed
result = guard.score_chain(question, steps, final_answer)

print(result.risk_score)        # float [0,1]
print(result.confidence_tier)   # "HIGH" / "MEDIUM" / "LOW"
print(result.needs_alert)       # True when risk >= 0.70
print(result.failure_mode)      # "retrieval_fail" / None / ...
print(result.behavioral_score)  # SC_OLD score before any blending
```

3.3 AgentGuard + P(True) — Cross-Domain Scoring

What it is: Behavioral SC_OLD blended with a P(True) signal from Haiku. P(True) prompts the model to rate answer correctness 1–5, providing a domain-agnostic uncertainty signal.

When to use: Cross-domain deployment where the target domain differs from the training distribution. Adds ∼$0.0003/chain.

Performance: AUROC 0.78 cross-domain (best ensemble, exp120).

```
from llm_guard import AgentGuard

guard  = AgentGuard(api_key="sk-ant-...")
result = guard.score_with_ptrue(question, steps, final_answer,
                                ptrue_weight=0.9)

# Or use the adaptive bandit to tune ptrue_weight automatically
guard.enable_ptrue_bandit()
result = guard.score_with_ptrue(question, steps, final_answer)
```

3.4 AgentGuard + Sonnet Judge — Highest Precision

What it is: Full J5 ensemble: SC_OLD ×1 + Sonnet judge ×3. The Sonnet judge produces GOOD/BORDERLINE/POOR chain-level verdicts.

When to use: Compliance-sensitive pipelines, high-stakes answers, or when precision matters more than cost.

Performance: AUROC 0.777 within-domain, 0.741 cross-domain (exp89).

```
from llm_guard import AgentGuard

guard  = AgentGuard(api_key="sk-ant-...", use_judge=True)
result = guard.score_chain(question, steps, final_answer)

print(result.judge_label)   # "GOOD" / "BORDERLINE" / "POOR"
print(result.risk_score)    # J5 = SC_OLD*1 + judge*3 (normalised)
```

3.5 stream_guard — Mid-Chain Abort

What it is: Real-time abort decision evaluated at step 2 of the agent loop. Combines SC_OLD prefix scores with an optional Haiku mid-chain judge call.

When to use: Long-running chains where early abort saves 3–5 LLM calls. Particularly effective when empty observations appear at step 1–2.

Performance: Two-stage OR strategy achieves Recall 0.416, Precision 0.740 (vs J5-only: Recall 0.595, Precision 0.908). Use when recall matters.

```
from llm_guard import AgentGuard

```

```
guard = AgentGuard()     # or AgentGuard(api_key=...) for Haiku judge

# Call after step 2 completes
sg = guard.stream_guard(question, steps_so_far, abort_threshold=0.65)

if sg.abort:
    print(f"Abort at step {sg.step_index}: risk={sg.risk_at_step:.3f}")
    # sg.rewritten_queries contains 3 diverse reformulations (if api_key set)
    next_query = sg.rewritten_queries[0] if sg.rewritten_queries else question
```

3.6 A2ATrustObject — Multi-Agent Trust

What it is: Structured confidence envelope passed between agents. HMAC-SHA256 signed over 8 canonical fields (exp114). Google A2A compatible.

When to use: Multi-agent orchestration. Agent A emits a trust object; Agent B conditions its strategy on the confidence tier and failure mode.

Performance: Round-trip pass 200/200 (100%); tamper detection 200/200 (100%); sign latency 0.034 ms; wire size 458–539 bytes.

```
from llm_guard import AgentGuard, A2ATrustObject

guard = AgentGuard()
trust = guard.generate_trust_object(question, steps, final_answer)

# Agent A: sign and transmit
trust.sign("fleet-secret-2026")
payload = trust.to_dict()    # JSON-serialisable dict

# Agent B: deserialise and verify
trust = A2ATrustObject.from_dict(payload)
assert trust.verify("fleet-secret-2026")  # raises if tampered

if trust.should_rewrite:
    # Use QueryRewriter before running Agent B
    pass
```

3.7 QuickCalibrator — Domain Calibration

What it is: One-call domain-adaptive isotonic calibration. Wraps `calibrate_isotonic()` into a 20-chain setup flow.

When to use: Cross-domain deployment where default thresholds give too many false positives or misses. Needs 20+ labeled chains from the target domain.

Performance: Isotonic calibration improves threshold precision without changing AUROC (monotone transform). Particularly effective for domains with different base rates of failure.

```
from llm_guard import QuickCalibrator

cal = QuickCalibrator(min_chains=20)
cal.fit(labeled_chains, domain="my_customer_support_bot")
# labeled_chains: each dict has "question", "steps",
#                 "final_answer", "correct" (bool)

risk = cal.score(question, steps, final_answer)
```

```
# Returns calibrated risk in [0,1]
```

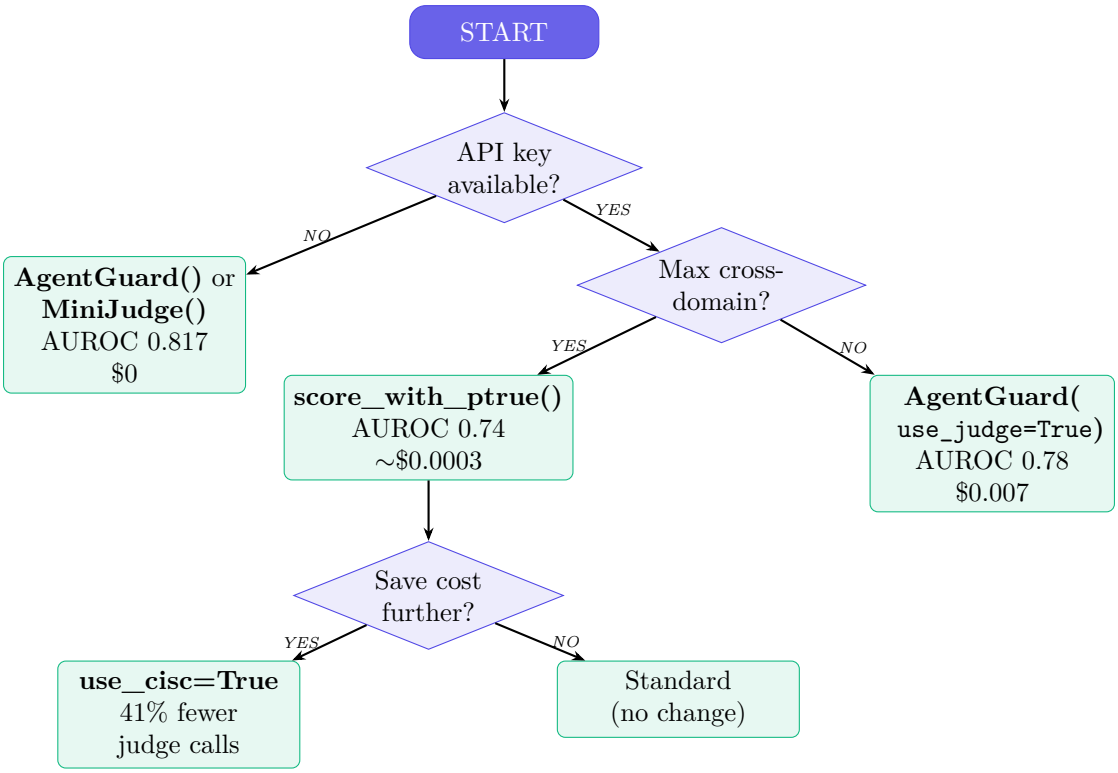
3.8 Platform Integrations

`llm-guard-kit` ships 2-line drop-in callbacks for major frameworks:

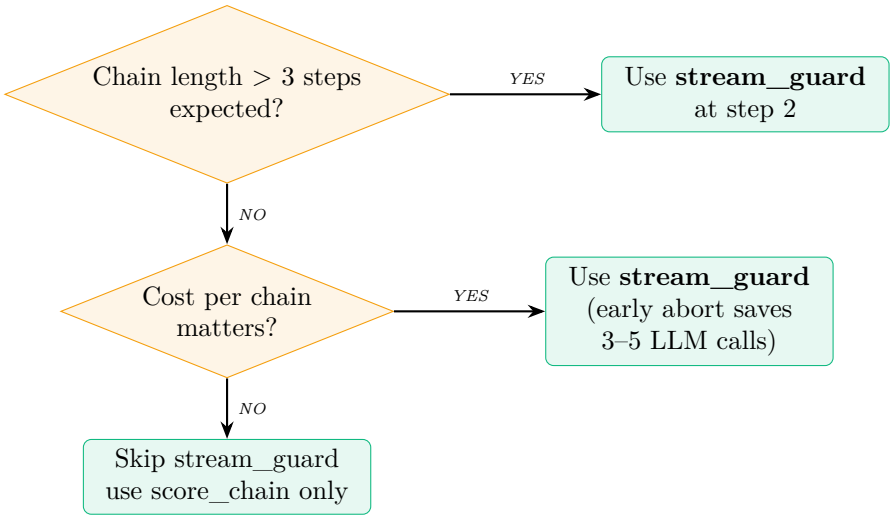
```
# LangChain
from llm_guard import AgentGuardCallback
from langchain.agents import AgentExecutor

agent = AgentExecutor(agent=..., tools=...,
                      callbacks=[AgentGuardCallback(api_key="sk-ant-...")])

# LlamaIndex
from llm_guard import AgentGuardEventHandler
from llama_index.core import Settings
Settings.callback_manager.add_handler(AgentGuardEventHandler())

# CrewAI
from llm_guard import AgentGuardCrewCallback
crew = Crew(agents=[...], tasks=[...],
            callbacks=[AgentGuardCrewCallback()])
```

4 Decision Guide

Use the flowchart below to select the right configuration for your use case.



When to use `stream_guard`:



5 Performance Benchmarks

All AUROC figures are from held-out evaluation sets; no in-sample figures are reported. Confidence intervals are computed via bootstrap (1000 resamples).

Method	Dataset	n	AUROC	95% CI	Cost/chain
SC_OLD behavioral	HotpotQA (within)	200	0.817	[0.74, 0.89]	$0
MiniJudge (LogReg+SC_OLD)	HotpotQA 5-fold CV	200	0.747 ± 0.10	—	$0
SC_OLD	TriviaQA (cross)	1000	0.659	[0.614, 0.705]	$0
SC_OLD	2WikiMultiHop (cross)	200	0.703	[0.628, 0.775]	$0
SC_OLD	NQ open-domain	200	0.524	[0.447, 0.609]	$0
+P(True) Haiku	TriviaQA	37	0.720	—	~$0.0003
+P(True) blend $\alpha = 0.25$	TriviaQA	37	0.737	—	~$0.0003
SC_OLD+Sonnet (J5)	HotpotQA	200	0.777	—	~$0.007
SC_OLD+Sonnet (J5)	TriviaQA	200	0.741	—	~$0.007
Alert precision @ 0.70	HotpotQA	200	—	P=0.908	$0
CISC cost reduction	HotpotQA	200	identical	—	−41%

Table 2: Full benchmark table. All results from held-out evaluation only. NQ open-domain AUROC 0.524 reflects the harder factoid domain gap; behavioral features are format-specific and degrade on truly unseen distributions.

6 Integration Patterns

6.1 Plain Python (simplest integration)

```python
from llm_guard import AgentGuard

guard = AgentGuard()    # $0, no API key needed

# After your agent finishes
result = guard.score_chain(question, steps, final_answer)
if result.needs_alert:
    print(f"WARNING: risk={result.risk_score:.3f}, "
          f"mode={result.failure_mode}")
```

6.2 Langfuse (trace + score)

```python
from langfuse import Langfuse
from llm_guard import AgentGuard

langfuse = Langfuse()
guard    = AgentGuard()

def run_and_score(question, run_agent):
    with langfuse.trace(name="agent_run") as trace:
        steps, final_answer = run_agent(question)
        result = guard.score_chain(question, steps, final_answer)
        trace.score(
            name="llm_guard_risk",
            value=result.risk_score,
            comment=f"tier={result.confidence_tier}, "
                    f"mode={result.failure_mode}",
        )
    return final_answer, result
```

6.3 LangSmith (evaluator protocol)

```python
from langsmith.evaluation import EvaluationResult
from llm_guard import AgentGuard

guard = AgentGuard()

def llm_guard_evaluator(run, example):
    question     = example.inputs["question"]
    steps        = run.outputs.get("steps", [])
    final_answer = run.outputs.get("final_answer", "")
    result       = guard.score_chain(question, steps, final_answer)
    return EvaluationResult(
        key="llm_guard_risk",
        score=1.0 - result.risk_score,   # higher = better
        comment=f"tier={result.confidence_tier}",
    )

# Then: client.evaluate(dataset_name,
    evaluators=[llm_guard_evaluator])
```

6.4 Prometheus (metrics + Grafana)

```python
from prometheus_client import Histogram, Counter, start_http_server
from llm_guard import AgentGuard

RISK_HIST  = Histogram("llm_guard_risk_score", "Risk score
    distribution",
                       buckets=[0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0])
ALERT_CTR  = Counter("llm_guard_alerts_total", "Number of
    alert-level chains")

guard = AgentGuard(on_alert=lambda r: ALERT_CTR.inc())

def score_and_observe(question, steps, final_answer):
```

```
    result = guard.score_chain(question, steps, final_answer)
    RISK_HIST.observe(result.risk_score)
    return result

start_http_server(8000)   # Prometheus scrapes /metrics
```

6.5 Datadog (DogStatsD)

```
from datadog import DogStatsd
from llm_guard import AgentGuard

statsd = DogStatsd()
guard  = AgentGuard()

def score_and_ship(question, steps, final_answer):
    result = guard.score_chain(question, steps, final_answer)
    statsd.gauge("llm_guard.risk_score", result.risk_score,
                 tags=[f"tier:{result.confidence_tier}",
                       f"mode:{result.failure_mode or 'none'}"])
    if result.needs_alert:
        statsd.increment("llm_guard.alerts",
                         tags=[f"mode:{result.failure_mode or
                             'none'}"])
    return result
```

7 API Quick Reference

7.1 Public Methods

Class	Method	Parameters	Returns
MiniJudge	score()	question, steps, final_answer	float [0,1]
MiniJudge	score_chain()	question, steps, final_answer	float [0,1]
MiniJudge	fit()	chains, n_splits=5	MiniJudge
AgentGuard	score_chain()	question, steps, final_answer	ChainTrustResu
AgentGuard	score_with_ptrue()	question, steps, final_answer	ChainTrustResu
AgentGuard	generate_trust_object()	question, steps, final_answer	A2ATrustObject
AgentGuard	stream_guard()	question, steps_so_far, abort_threshold	StreamGuardRe
AgentGuard	activate_adapter()	failure_mode	AdapterResult
AgentGuard	route_to_mesh()	question, trust, agent_answers	MeshResult
AgentGuard	calibrate_isotonic()	scores, labels	AgentGuard
A2ATrustObject	sign()	secret	A2ATrustObject
A2ATrustObject	verify()	secret	bool
A2ATrustObject	to_dict()	—	Dict
QuickCalibrator	fit()	chains, domain	QuickCalibrator
QuickCalibrator	score()	question, steps, final_answer	float
probe_ensemble_blend()	(module fn)	probe_score, ptrue_score, alpha=0.25	float

Table 3: All public API methods. `steps` is always a list of ReAct step dicts with keys: `thought`, `action_type`, `action_args`, `observation`.

7.2 ChainTrustResult Fields

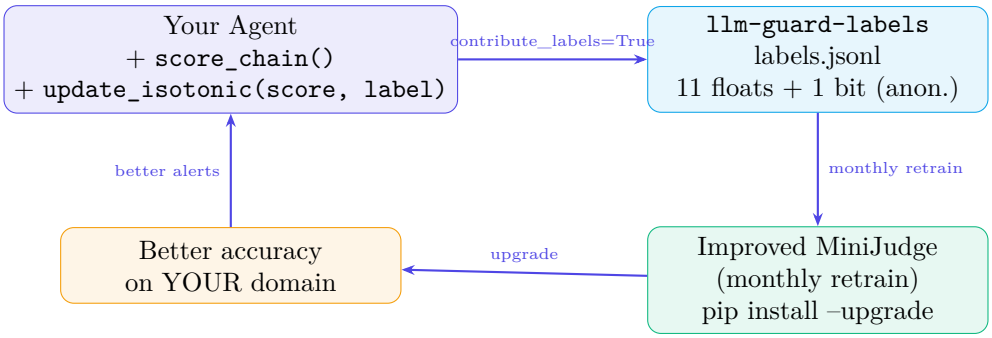
Field	Type	Description
`risk_score`	float [0,1]	Composite risk (higher = more likely wrong)
`confidence_tier`	str	`"HIGH"` / `"MEDIUM"` / `"LOW"`
`needs_alert`	bool	True when risk_score ≥ alert_threshold (0.70)
`failure_mode`	str|None	`"retrieval_fail"` / `"repeated_query"` / `"long_chain"` / `"empty_answer"` / None
`judge_label`	str|None	`"GOOD"` / `"BORDERLINE"` / `"POOR"` (requires `api_key` + `use_judge=True`)
`behavioral_score`	float	SC_OLD score alone
`behavioral_components`	Dict	Individual SC_OLD feature scores (sc1–sc12)
`latency_ms`	float	Scoring latency in milliseconds
`step_count`	int	Number of Search steps in the chain

Table 4: Fields of `ChainTrustResult` returned by `score_chain()`.

8 Telemetry Flywheel

8.1 Architecture

The telemetry flywheel allows users to contribute anonymised labels that improve the shared MiniJudge model over time.



8.2 Privacy Guarantee

Only the following data is transmitted when `contribute_labels=True`:

- **11 float features** (SC_OLD structural scores: loop rate, empty-observation rate, step count, etc.)
- **1 correctness bit** (0 = correct, 1 = wrong)

No question text, no answer text, no user data, no PII is transmitted. The 11 features are computed locally and contain no recoverable content. Label contribution is **opt-in only** and disabled by default.

```python
from llm_guard import AgentGuard

guard = AgentGuard()
result = guard.score_chain(question, steps, final_answer)

# After you know the ground truth:
```

```
guard.update_isotonic(result.behavioral_score, label=1)
# label=0 means correct, label=1 means wrong

# Opt-in to anonymised label contribution (default: False)
guard.contribute_label(result, correct=False,
                       github_pat=None)   # set PAT to contribute
```

Quick Start

```
pip install llm-guard-kit

from llm_guard import AgentGuard

guard = AgentGuard()   # zero cost, no API key

# Score a completed agent chain
result = guard.score_chain(
    question     = "What year was the Eiffel Tower built?",
    steps        = [
        {"thought": "Searching...", "action_type": "Search",
         "action_args": "Eiffel Tower year",
         "observation": "Built 1887-1889, completed for 1889 World
            Fair."},
        {"thought": "Got it.", "action_type": "Finish",
         "action_args": "1889", "observation": ""},
    ],
    final_answer = "1889",
)

print(f"Risk: {result.risk_score:.3f}  Tier:
   {result.confidence_tier}")
# >> Risk: 0.133  Tier: HIGH

if result.needs_alert:
    print("ALERT: possible wrong answer")
```

PyPI: https://pypi.org/project/llm-guard-kit/ • **GitHub:** https://github.com/Avighan/llm-guard-kit